

Degrowth and (un)sustainable lifestyles in two Danish ecovillages

Peter Nielsen¹  and Lone Riisgaard² 

Abstract

Based on an extensive qualitative study, this paper explores sustainability in two Danish ecovillages and discusses the findings (meta)theoretically. Linked to the degrowth literature, it takes its point of departure in critical realism by identifying mechanisms related to (un)sustainability in these ecovillages. The most important mechanism generating their sustainable practices is identified as a subsistence political economy, a mechanism rooted in infrastructures such as regenerative farming and direct democracy. However, there is another important mechanism affecting the sustainability of the ecovillages, namely middle-class ideals. This mechanism generates unsustainable, growth-oriented lifestyles, which are also rooted in infrastructures such as large family houses and divisions between the private and the communal. Generally, it is concluded that in these two ecovillages the subsistence political economy mechanism ensures a move towards more sustainable lifestyles for all villagers, while infrastructure sustaining middle-class ideals pulls in the opposite direction.

Keywords

Degrowth, critical realism, ecovillages, qualitative methods, sustainability

¹Department of Social Sciences and Business, Roskilde University, Roskilde, Denmark

²Department of Social Sciences and Business, Roskilde University, Roskilde, Denmark

Corresponding author:

Peter Nielsen, Department of Social Sciences and Business, Roskilde University, Universitetsvej 1, 4000 Roskilde, Denmark.

Email: petern@ruc.dk

Introduction

Whereas Denmark is often portrayed in political and public discourse as a green front-runner or a shining example of progressive climate policy, recent critical studies paint a rather different picture. Tilsted and Bjørn (2023) evaluated Danish climate policy targets and found that ‘cumulative allowable emissions that follow from different equity approaches reframe the Danish narrative from a green frontrunner to an indebted culprit’ (Tilsted and Bjørn, 2023: 15). In particular, recognising that past emissions of CO₂ should also be accounted for highlights the question of Denmark’s mounting climate debt (see also Vogel and Hickel, 2023).

Although Denmark’s national climate policies are failing to do their fair share of mitigating the ongoing ecological catastrophe, local-level and civil society-based strategies are seeking to reverse this trend. The ecovillages on which we focus in this paper are examples of this.

As we shall argue below, the existing degrowth literature on ecovillages either tends to be celebratory, highlighting their position as frontrunners of degrowth and sustainability, or more rarely dismisses them as examples of idealised localism. In fact, transformative practices are rarely all good or all bad (Buch-Hansen and Nesterova, 2023). On the contrary, our empirical data show sustainable and unsustainable practices co-existing in the same villages. Hence, answering a gap in the existing scholarly literature on ecovillages, in this article we develop a grounded and nuanced understanding of their sustainability by identifying the mechanisms which cause ecovillage lifestyles to be unsustainable or sustainable.

Taking its starting point in the degrowth literature, and utilising critical realist philosophy of science, this study proceeds as follows. The first section below reviews the existing literature on degrowth and ecovillages. The following section introduces our approach to degrowth and sustainability in the light of critical realism. The third section describes the methods used in our qualitative study, carried out in two Danish ecovillages. The fourth and fifth sections respectively analyse the empirical data and identify mechanisms and agency related to (un)sustainable lifestyles in the ecovillages. The penultimate section contains (meta)theoretical reflections, relating the study to the conceptual work of Brand and Wissen (2021), while the last section sums up and discusses the empirical and analytical findings.

Degrowth and ecovillages

Degrowth consists primarily of a concept and a discourse, articulated by academics and activists, that focuses on the detrimental social and ecological effects of economic growth while at the same time depicting favourable alternatives that emphasise sustainability, well-being and direct democracy. At least, this is one way of pinning down an increasingly complex and self-reflective approach (Buch-Hansen et al., 2024; Schmelzer et al., 2022; Vandeventer and Schmid, 2024; Windegger and Spash, 2023). Much of the writing on degrowth is theoretical and conceptual, considering practical matters only indirectly by proposing top-down policies combined with bare mentions of

bottom-up grassroots and social niche experiments (Polewsky et al., 2024). Along these lines, the bulk of the degrowth literature looks favourably on ecovillages, as is the case, for instance, with seminal work by Serge Latouche and Tim Jackson. Latouche (2009) highlights *re-localisation* as one of the most important goals in developing a degrowth society. He envisions local autonomy as a node in a network of solidaric relations that act as both laboratories for democratic regeneration and bulwarks against growthism. While we await the necessary steps on the national and global level, Latouche (2009) writes, we can look to a multitude of local initiatives, rural or urban, such as ecovillages for inspiration. Moreover, Latouche (2009) also calls for a variety of structural changes in a degrowth society such as a reduction in working hours and the restoration of sustainable farming, as well as of taxes on financial transactions and CO₂ emissions (see also Kallis et al., 2020). Along the same lines, Jackson (2009) points to numerous small-scale initiatives at the grassroots level, such as ecovillages that go against the treadmill of material growth and experiment with more ecologically sustainable lives with an emphasis on promoting well-being. However, he also points to the fact that such developments remain quite marginal and are severely constrained by structural factors. Such local acts of resistance are ‘destined to failure’ (Jackson, 2009: 152) without structural changes in society. Jackson (2009) points specifically to lessons from ecological economics, such as resource and emissions caps, and revising the national accounts to go beyond GDP as the measure of progress.

Framed by such influential work, the degrowth literature on ecovillages contains a main thread that operates mainly at the structural and theoretical level, but that sees ecovillages in a positive light as a vital local source of small-scale inspiration and experimentation. However, we also find theoretical accounts of degrowth that go against this tendency, as illustrated below. Importantly, moreover, there are also empirical studies on degrowth and ecovillages that mirror the divide in the theoretical debate by either underlining highly positive findings or (less frequently) focusing on more problematic issues.

Although degrowth and ecovillages are distinct concepts they are frequently combined to put flesh on more abstract degrowth literature. Writings on degrowth and ecovillages that adopt a general and theoretical nature highlighting ecovillages as an illustration of degrowth are common. Cattaneo (2015: 167) notes that the ecovillage ‘provides insights as to how a degrowth society might look’ and add that eco-villagers become the authors of their own story and ‘ferment a degrowth imaginary’ by placing the community at the center of life, rather than capitalist or state authority (see also Buch-Hansen, 2019; Fonseca et al., 2022). In a recent introduction to the notion of degrowth, Schmelzer et al. (2022) points to the importance of interstitial strategies that seek to experiment with new institutions, infrastructures and organisational forms. Such strategies are relevant in so far as they show that the ‘principles of a degrowth society are already being implemented on a small scale today’ (Schmelzer et al., 2022: 256; see also Trainer, 2006, 2012).

Sticking for now to the conceptual and theoretical writings on degrowth, we also find a few examples that go directly against the stream by debunking the view that ecovillages and degrowth fit together like hand in glove. Xue (2014) challenges the view that

localisation and decentralisation necessarily lead to a just and sustainable degrowth society. She points to the thermal inefficiency of single-family houses that is common in ecovillages and also argues that ‘the low-density and scattered distribution of villages will result in longer travel distance and use of [a] motorized transport mode’ (Xue, 2014: 133). As regards a grand transition to ecovillages, Xue finds that this would result in a waste of resources invested in current infrastructure, a large increase in built-up areas, and a rebuilding and adaptation process that ‘is by no means environmentally neutral’ (Xue, 2014: 133). Following such observations, she proposes instead compact cities as the best small-scale example of degrowth, not least since such dwellings are also more open-minded than what she considers to be the typically uniform ecovillage (see also Mocca, 2020). Buch-Hansen and Nesterova (2023) also identify a tendency in the literature to ‘make degrowth sound like a wellness retreat whereby downscaling is a process of descent into a state of wellbeing, unfolding in idyllic, mostly localised, cooperative communities’ (Buch-Hansen and Nesterova, 2023: 4). They point out that whereas for some, living in ecovillages and engaging wholeheartedly in community-based activities might improve well-being, others might live out degrowth and flourish in different ways (see also Buch-Hansen et al., 2024: 66–7, 131).

Turning now to empirical scholarship on degrowth and ecovillages, a strand of literature likewise finds a close match and other studies are more sceptical. Starting out with the former strand, Litfin (2014) conducted an empirical study of fourteen ecovillages around the world, finding all in all that ecovillages exemplify the inevitability of degrowth by internalising the fact that infinite growth on a finite planet is impossible; thus ‘we can look to ecovillagers as our forerunners’ (Litfin, 2014: 196). Lockyer (2017) studied the Dancing Rabbit Ecovillage in the USA and found that ‘material degrowth is achievable without significantly sacrificing quality of life’ (Lockyer, 2017: 538). He concludes that in places like Dancing Rabbit, adherents of degrowth ‘have found our muse’ (Lockyer, 2017: 539). In a Danish context, Verco (2019) writes about the eco-community – or Freetown – of Christiania and argues that this eco-community is ‘an outstanding and long-standing oppositional case’ for degrowth proponents (Verco, 2019: 106). In another Danish case study, Gausset and Jensen (2024) analysed the Self-Sustaining Village and concluded in support ‘of the degrowth argument that another, better, and happier world is possible with lower levels of consumption and resource exploitation’ (Gausset and Jensen, 2024: 4).

Empirical studies of ecovillages that arrive at more critical conclusions concerning degrowth are few and express themselves more indirectly. Marckmann et al. (2012) conducted a case study of Danish ecovillages and their empirical data suggest that the values of ecovillage residents do not necessarily diverge from the mainstream. The authors conclude that ecovillagers do not challenge the expectation of a constant growth in consumption (Marckmann et al., 2012: 428). Temesgen (2020) studied ecovillages in Norway and Scotland with a particular interest in the consequences of their physical expansion. In one village this process led to the ecovillage being marketed with ‘fidelity to the mainstream capitalistic worldview’ (Temesgen, 2020: 20). Looking at both cases, she concludes that ‘mainstream values seem to infiltrate the ecovillages’ (Temesgen, 2020: 20).

Reviewing the literature on degrowth and ecovillages, we can note that, even though the resulting image is diverse, there is a strong tendency to confirm the compatibility

between degrowth and ecovillages theoretically or even celebrate the exemplary properties of ecovillages as frontrunners of degrowth. The empirical literature largely corresponds with that tendency. Even though critical studies exist, they are few. There also seem to be two rather constant binaries in that either the work is theoretical, or it is empirical, and either it is celebratory, or it is dismissive.

Our empirical findings corroborate many of those found in both the celebratory and the dismissive literature, pointing to a need for an analytical approach which can encompass and explain such diversity. Hence, the present work contributes to the degrowth literature by going beyond binaries and proposing a nuanced analysis utilising the critical realist philosophy of science. Rather than making sweeping generalisations, the present study uncovers the complexity of real life and experiences in the ecovillages and hence contributes to existing discussions about whether and how ecovillages contribute to a degrowth transformation.

A critical realist approach to sustainability

Critical realism is a philosophy of science perspective that originated in the 1970s as an alternative to positivism and has also been articulated in later decades as an alternative to radical constructivism (Buch-Hansen and Nielsen, 2020). Whereas most existing work on critical realism and degrowth are *conceptual* in nature (Buch-Hansen et al., 2024; Buch-Hansen and Nesterova, 2021; Buch-Hansen and Nielsen, 2020, 2023), the present study *applies* critical realism.

In critical realism, the essence of social science as an overall research design is the movement from knowledge of empirical phenomena to knowledge of the structures and mechanisms that generate or sustain these phenomena (Bhaskar, 2015; Buch-Hansen and Nielsen, 2020; Fletcher, 2017). *Structures* are real patterns with specific powers and liabilities enabling *mechanisms* to work in particular ways and to cause events and phenomena by means of human *agency* (Buch-Hansen and Nielsen, 2020).

In other words, critical realism equates social science with combining empirical data, structures, mechanisms and agency to acquire deep knowledge of the research issue at hand, for instance working with abstraction relating to existing theories such as degrowth theories in this case. The methodology of abstraction allows the researcher to go beyond (or below) empirical data in order to focus on key unobservable features of social life (Buch-Hansen and Nielsen, 2020). However, in critical realism knowledge is never absolute since firstly, the social world changes, and secondly, all knowledge is fallible and lacks unquestionable foundations due to knowledge and social reality being two fundamentally different things. On the one hand, critical realism is ontologically realist and includes a deep and unobservable ontological domain, but on the other hand it is epistemologically relativist. This means that in critical realism no certainty is involved in science. We can never be sure that we are right and that calls for a humble approach. Mechanisms are not directly empirically verifiable or falsifiable, and interpretations of empirical data are dependent on a variety of social factors and theoretical preconditions, conscious or not. Casting reality and scientific discourse as two basically different

dimensions and following epistemic relativism means that all scientific work is wide open for developments.

Mediating between realism and relativism, critical realism involves judgmental rationality, i.e. reflecting on the merits of competing interpretations of social reality, as intrinsic to the scientific endeavour (Bhaskar, 2008, 2011; Buch-Hansen and Nielsen, 2020). Critical realism entails a qualitative focus in social science and prioritises conceptual precision and interpretative dialogue over measurement and quantitative accuracy (Bhaskar, 2015). This latter point is particularly relevant to our approach to sustainability since there are numerous ways to approach this issue.

As with other concepts containing positive connotations, sustainability is a contested concept (Buch-Hansen, 2019; Buch-Hansen and Nielsen, 2024; Hoff et al., 2020; Næss, 2006). While recognising how difficult it is to agree on a definition, we do not consider sustainability to be an empty signifier. We also affirm that even though degrowth is not only about sustainability it would be devoid of meaning without relating strongly to sustainability. Working with degrowth and critical realism, we consider sustainability to be an ethical idea entailing that existing human production, consumption and social life should respect both ecological limits and principles of equality and solidarity. Sustainability is a holistic approach linking political economy, human needs and nature, and it means that principles of care and solidarity are to be respected globally and for all generations (Buch-Hansen and Nielsen, 2020: 94; Parker, 2007). In line with critical realism and degrowth, sustainability entails that consumerism and aggregate economic growth must end in the Global North and that capitalism must be replaced by a new system that is centered on ecology and equality (Buch-Hansen and Nielsen, 2020: 102–103; Næss, 2006).

Even though quantitative measures and indicators such as carbon footprints, ecological footprints, GDP, economic inequality, and self-reported life-satisfaction are relevant to sustainability (see, for instance, Daly, 2017; Gausset and Jensen, 2024; Jackson, 2009), in our critical realist approach, sustainability relates to structures, mechanisms and actual lifestyles and hence is better captured using a qualitative approach. While embracing a holistic understanding of sustainability, we focus our analysis on the mechanisms and structures, which came out as most strongly related to (un)sustainability in our empirical data. These include the interrelations between political and economic life on the one hand and ecological sustainability on the other hand. When referring to political economy in this context, we mean to emphasise the intricate interconnections of politics and economy in the ecovillages so that decision-making, values, organisation, work, distribution and consumption are woven together to form complex empirical modes of political economy supported by (infra)structures and mechanisms. The concept of infrastructures in this case refer to both *physical* infrastructures such as housing and heating and to *social* infrastructures such as direct democracy and eating in common.

Data and methods

The empirical data in this study were obtained by the authors from two anonymised ecovillages in Denmark. The first (hereafter called EV1) is located adjacent to a small

‘ordinary’ village in a rural area around 10 km from the nearest train station. EV1 is populated by about seventy people, of whom around half are children, and it recently celebrated its twentieth anniversary. The residents build their own houses or buy houses built by previous residents. Only a few have resided there for the full twenty years. The other village (hereafter called EV2) is located close to a small town with a train station. It is populated by about 220 people, of whom around a third are children. The accommodation consists of ninety houses that are alike albeit of slightly different sizes. They were built together before the first residents moved there in 2018. Half of the houses are owned or co-owned by the residents, whereas the other half are let for rent. Since its inception, around one third of the original residents have moved out, and new residents have moved in. Both EV1 and EV2 have open spaces around old farms (part of them renovated), which form centers of community life and contain communal facilities such as machinery and tool storage and guest rooms.

The villages were selected to provide variety not only between a larger and a smaller community but also a newer and a more established one. The logic being that if similar mechanisms and structures were identified across these differences, the propositional value of our analysis would strengthen. The data were collected as part of an international research project on the interconnections between direct democracy, community and green transition in which the research had a participatory element. Therefore, before deciding on the precise research questions, we engaged in informal conversations with residents of both ecovillages, and it is from these conversations that the research question in this study originates. When asked what it would be interesting for them to know more about, one of the questions a resident pinpointed was whether people in the ecovillage are actually living sustainably. Her own response was that they are not, and this sparked our interest as researchers. We found that the concept of degrowth is not commonly used in the ecovillages whereas the concept of sustainability certainly is, and due to the participatory element of our study we are employing both concepts in this paper.

Other than a broad grounding in degrowth scholarship, no theoretical presuppositions preceded collecting the data, and to acquire a broad picture of life and attitudes in the ecovillages, we decided on a wide spectrum of qualitative methods. Over a one-year period in 2023/24 we conducted 27 semi-structured interviews across the two ecovillages spanning residents of different ages, gender, family and work circumstances, lifestyle and number of years spend living in the ecovillage. We also engaged in making observations, including participant observations. Regarding the former, we observed a variety of meetings and undertook extended field visits to both ecovillages spanning several days to observe and experience everyday life. Regarding the latter, we participated in a variety of activities in the villages, such as cooking communal meals, weeding, harvesting and preparing new beds for regenerative farming.

In addition, the researcher who collected data in EV1 has lived in EV2 since 2020. Therefore, data in EV2 were obtained by the other researcher. This lived engagement however ensured a bridge between embodied experience and theoretical discussions which enriched the analysis. Methodologically this can be construed as a combination of long-term participant observation (Gausset and Jensen, 2024) and a form of activist research (Lockyer, 2017).

Identifying structures and mechanisms

In this section, we present and analyse empirical data in order to identify structures and mechanisms related to (un)sustainable lifestyles in the ecovillages by means of abstraction. Working with critical realism entails reasoning that moves from the empirical to the deep (unobservable) domain of reality and back again in a creative iterative process of abstraction and concretion. We do this by focusing mainly on the political economy of everyday life in the villages and we start by describing the villages empirically and presenting structures promoting sustainable lifestyles leading to the identification of the mechanism of subsistence political economy that these structures support. This is followed by identification of a second mechanism of middle-class ideals promoting unsustainable behaviour in village life.

Decision-making

To some extent, the political economies of EV1 and EV2 are the same. Starting out with what is usually referred to as politics, in both ecovillages we observed a vibrant direct democracy in action revolving around bottom-up participation. Both ecovillages have a dual overall organisation where on the one hand the land and other communal assets are owned collectively by a company where the main responsibility is economic viability. On the other hand, most of what goes on from day to day revolves around democratic structures. Aspiring to achieve a high degree of self-sufficiency when it comes to food, organic or regenerative farming both in greenhouses and on open land are instrumental in both ecovillages. In EV1, the modes of decision-making are quite loose on the ground. Farming and other community activities take place in teams like the vegetable team and the bread team, which are autonomous but operate within budgetary constraints and respect the decisions of the community at large. The latter are reached through regular, usually monthly communal meetings in which around a third of the adult residents participate. The democratic ideal is consensus, so that no decision is made until no opposition is voiced. This can be a lengthy process. For instance, we observed that a major decision about the future of the old farmhouse could not be reached at a particular meeting because of differing opinions.

Being around four times bigger, measured by the number of adult residents, EV2 has a more layered decision-making structure also involving representation and occasional majority rule. The basic units in EV2 are also rather autonomous teams, of which there are many, such as the harvesting team, the bee team, the willow team, the strawberry team, the machinery team and the technical team. Each year there is a process where the adult residents can switch teams. New teams are also common, and most residents participate in one or two. The teams are organised in areas such as an area for farming and have one or two representatives participating in the decision-making in their area. All areas then send one or two representatives to a 'general council' that meets regularly to arrive at larger decisions. Teams, areas and the council all work in confluence with decisions arrived at through the regular, usually monthly general meetings for all residents, which are typically attended by around a quarter of the adult population.

In both ecovillages, meetings are carefully planned and led by particular teams so as to facilitate good decisions by raising awareness, considering reservations and opposing views, and avoiding bad communication, personal conflicts and never-ending meetings. In EV2 this team is part of an area termed social sustainability, which seeks to ensure efficient and consent-based decision-making. Such procedures are very important for the cohesion of the ecovillages and their progress. The functioning of political life in the ecovillages owes a lot to the way the concrete process of decision-making is handled and carried out. We found this to be a result of a learning process where in both ecovillages, previous conflicts and bad communications have sparked the need for change. For instance, one person from EV1, who moved there about ten years ago, remarked in an interview that, ‘when I first came here the community were not so good at communicating. That has improved a lot. [...] Sustainability is also about social sustainability – about how to interact with one another’. Similar developments were noted in EV2, as illustrated in the following quote: ‘Well, you have to accept each other completely as we are – for it to be sustainable, if I may say so’.

The physical infrastructure

Apart from the local organic or regenerative production of food, both ecovillages also use other environmentally friendly techniques. Houses are built from environmentally friendly materials, and renovations of old buildings are handled with sustainability in mind. Houses are well insulated and consume relatively low amounts of energy. Heating is based on non-fossil-fuel energy in both ecovillages. However, while in EV1 the method of heating is different for each house, EV2 has a general infrastructure with geothermic heating run by a windmill. The windmill also provides energy for communal facilities such as the communal kitchen and washing machines and powers electric cars. Both villages also have a willow wastewater-cleaning facility where waste from sinks and toilets is run through a system drawn from the idea of permaculture, in which a circular system uses wastewater to grow fast-growing willow trees. These trees are then regularly cut down and in EV1, for instance, are for heating purposes or in EV2 made into woodchips which are spread in the beds devoted to regenerative farming. In EV2, where the physical infrastructure is most advanced, there is also a rainwater collection system where rainwater from the roofs of the houses is stored and used in toilets and washing machines.

Work and consumption

In both EV1 and EV2, observed communal economic life revolves around voluntary and unwaged labour performed in the teams that members choose to join. No general accounts of hours spent are available in the villages, and in EV1 the work is strictly voluntary, whereas in EV2 all adults are expected to be a member of at least one team and have responsibilities that amount to around two hours a week on average. However, we observed that most residents work more, some a lot more, this being necessary for the overall economic viability of the village, but also causing occasional irritation and

frustration. In both villages, most members have paid work outside the village or have economic means to sustain themselves being for example pensioners. In EV1 everything produced on communal land is available to all, and other basic food and household necessities such as flour, butter, milk, beans, soap and toilet paper are also bought and stored, being made available for all according to need in the old farmhouse. The costs are covered by monthly payments from the residents, around 1200 DKK/ €160 a month for an adult, which also includes communal meals, where residents take turns in cooking and serving dinner for the others. Such meals are held four times a week in EV1, are mostly vegetarian, and as far as possible use what is produced in the village. In summertime, the local produce covers a lot of the local needs, though this is less the case in the rest of the year.

In EV2, communal meals take place every weekday except for school holidays, similarly being of great importance for local production and social interaction. In this village, the meals are also mostly vegetarian, with the option of a meat dish on one day a week. In both villages, the food can either be eaten in common or taken home. Participation in the communal meals in both EV1 and EV2 is voluntary and based on day-to-day lists. However, the economy of EV2 is quite different from that in EV1.

In EV2, communal meals are paid for individually, as is all the food that is produced on communal land. There is a small shop for the residents where the harvesting team put out food to buy, or else the residents can harvest vegetables for themselves. Everything is paid for item by item at prices usually slightly below the normal prices in the local supermarket. The economy of EV2 is different from EV1, mainly because a farmer and a part-time kitchen leader are employed by the village. This makes for a greater degree of self-sufficiency and the smoother functioning of the communal meals, but it also calls for a greater inflow of money. Not only do the villagers pay for the food produced in the village, but they also sell some of what is produced outside the village to sustain economic viability. In addition, the villagers make voluntary monetary contributions totalling around 500,000 DKK/ €67,000 annually, without which the village would not be economically viable.

The mechanism of subsistence political economy

Based on our empirical data, we can identify an unobservable mechanism in both villages that we term *subsistence political economy* (see Mies and Shiva, 2013). The concept of subsistence political economy is thus not a theoretical premise of this paper but a result of our analysis of the empirical data as it is found to signify a mechanism in the villages. This is a mechanism integrating political and economic life that is ecologically sustainable by being based on local organic or regenerative farming and other environmentally friendly practices. While this aspect is touched upon more briefly in the data presented in this article, it is also socially sustainable in that it encompasses equality and care, while respecting different needs and capabilities. It is not a political economy that relies on economic growth and being based primarily on unwaged labour in both the political and economic spheres, it does not generate high incomes that often translate into unsustainable levels of consumption. It values the preservation and regeneration of life in its different

forms, human and non-human. Thus, it is a mechanism that generates and recreates sustainable lifestyles in the villages in line with degrowth aspirations, and it is sustained by networks of both social and physical infrastructures, such as communal meals, direct democratic decision-making, circular wastewater treatment and non-fossil-fuel energy provision (see also Gausset, 2020; Gausset and Jensen, 2024; Nguyen, 2019).

The mechanism of middle-class ideals

However, the mechanism described above is not the only important one in relation to sustainability and lifestyles in the two ecovillages. Being interested in (un)sustainable lifestyles in the villages, this was one of the subjects of our interviews. Asked whether she considered the village to be sustainable, one resident from EV1 replied after hesitating:

I do not know why I cannot just say yes [...]. I know too well that we emit less CO₂ than the surrounding society. But I think that, for instance, all this about people building houses [...] that is really a lot of resources being used for building new houses. And very big houses, too.

Another reflective comment on the issue surfaced when the interviewer asked a resident of EV1 if there had been a change in her private consumption since she moved to the village:

I use the car for more and more things [...] I seldom use the bicycle anymore.

The tendency for some villagers to increase private transport by car after moving to the ecovillage was also evident in EV2, particularly related to the location of jobs and schools, but also linked to financial means. The following quote comes from a conversation about whether people in EV2 live sustainably:

I drive a diesel car because I can't afford to switch to an electric car. I can see over time that it has changed a lot at (EV2). I also have the ideology for it. I would also like to buy an electric car.

From our observations, we also know that cars abound in the villages and that they are being used for a variety of purposes and create a lot of traffic, most of which is fossil fuel-based. Some residents in both villages also fly, making unsustainable transport practices widespread.

Comments from villagers and our own observations led us by means of abstraction to identify a mechanism in both villages that we will refer to as *middle-class ideals*. These are ideals that can be seen as working against degrowth aspirations. They revolve around a nuclear family where a good life is thought of as including spacious accommodation, car(s), a second home for vacations, frequent flying and full-time wage labor. In other words, this is a consumer culture based on intensive work and high levels of consumption, as well as a growth-oriented culture including the accumulation of material goods

and private wealth. These ideals are unsustainable, as they are reproducing both inequality and ecological destruction.

In Denmark, these are the most widespread political-economic ideals and have been so increasingly since the middle of the preceding century, as higher wages and economic growth have made this lifestyle increasingly possible for the great majority of the population. Originating in the gestation of the consumer society in post-war welfare capitalism and the gradual withering away of the traditional working-class (Nielsen, 2011), these ideals are currently at the core of neoliberal capitalism and have been naturalised by mainstream economics (Buch-Hansen and Nielsen, 2024). As such, it is not only a mechanism presiding in the villages. However, it cannot be considered as emanating only from grand structures of mainstream society either, as it is also sustained by physical and social infrastructures that are particular to the ecovillages. In both ecovillages, roads and parking spaces make it possible to have cars close at hand, and in both ecovillages ownership of new-built spacious family houses is prevalent. On top of that, clear structural divisions are drawn between what is considered *communal* and what is considered *private* in the villages, the latter covering areas of life beyond communal rules and decisions.

The communal and the private

The communal form of political economy has mainly to do with affairs such as farming, meals, maintaining community relations, taking care of physical infrastructure and ensuring economic viability. This rather limited community leaves an open space for private dealings as well when it comes to both political and economic practices. We observed that villagers had quite different political leanings, ranging from neoliberal and market-friendly to communist or anarchistic. Some are vegetarians, others eat meat. Some are environmentalists and deeply concerned about the climate breakdown, whereas others are more interested in a community that is close to nature and a good place to bring up children and are attracted primarily by the community feeling. Regarding economic life, only a small part of the wealth and incomes of residents is channelled into the villages, whereas what remains is considered private, meaning that a major part of household consumption and investment are not matters of communal concern. Issues of private economic dealings come up in informal conversations, and questions of economic inequality also occasionally enter the democratic process, but all in all these matters are considered private and are generally excluded from the communal political-economic life. This leaves ample room for unsustainable lifestyles, which is generally viewed to be legitimate even when the values of simple living, minimalism and voluntary self-limitation are also present.

Our data from EV2 in particular suggest a deliberate normative standpoint on diversity and a desire to get many people to move in a sustainable direction, rather than having a few people living highly sustainably:

[...] I found out that people ate meat here. Which I was really shocked about. [...] It took me a long time to understand that in such a community everything is a compromise. [...] Well, I think

I just realized that it's better that we are 300 people who eat meat once a week - not that I do, but I accept that they do - than we all eat meat every day.

The aim for a more social form of sustainability rather than strict ecological sustainability on the household level is also explained with reference to the ability to function as a source of inspiration for people outside the village. A resident of EV2, for instance, remarked: 'It can become so avantgarde that you lose contact with society, and then the power of inspiration disappears. Then there is no path anyone can step on. Then all they can see is someone on an island'.

(Un)sustainable lifestyles and agency

Having identified two mechanisms in the previous section that exist *generally* in the villages we now turn to questions of actual lifestyles and agency in the villages. Thus, now we both return to questions of the empirical and bring in *agency* in order to explore actual lifestyles. Whereas the ecovillages in this study are not generally sustainable due to the prevalence and sustaining of middle-class ideals, there is great variety of lifestyles in the villages resulting from residents exercising agency in different ways. As regards (un)sustainability, the important question is to what extent the mechanisms are reproduced. In our empirical data, both mechanisms are always actualised in village lifestyles, but in very different combinations. Generally, and referring to the definition of sustainability and degrowth above, lifestyles that mainly reproduce middle-class ideals are unsustainable, but lifestyles that mainly reproduce the political economy of subsistence are more sustainable and in line with degrowth aspirations. Our empirical data show that it is possible to consciously seek to avoid middle-class ideals in the villages, and that some people actually do so, leading rather sustainable lives as a consequence. Some villagers are very much involved in the subsistence political economies of the villages by putting in many hours of unwaged labor and participating to a greater degree in community life, such as attending meetings, while at the same time they go against middle-class ideals by, for instance, not having a car, not flying, and consuming moderately. Thus, some people in both villages live as sustainably as one can reasonably expect, given that ecovillages and their residents are embedded in and rely on the larger productivist and often unequal and deeply unsustainable society for jobs, banking, financial systems, welfare remittals, communications systems and so on (see also Gausset and Jensen, 2024, Piketty, 2020; Xue, 2014).

There is also a very important dynamic at play here that relates to collective agency. It is not only lifestyles that can differ and change – collective agency can also change the physical and social infrastructures of the village. Such changes are imperative regarding sustainability over time.

Our empirical data contain examples of collective agency that work both for and against sustainability. One example of the latter is that in EV1 it is now commonplace to park one's cars right next to one's house, whereas earlier cars had to be parked in the parking lot by the entrance. Another example from EV1 is that individual households have begun to grow vegetables and herbs in their own gardens rather than on communal

land. Such changes, which amplify middle-class ideals, go against sustainability by further grounding an unsustainable car culture and hollowing out what is communal. An example pointing in the other direction from EV2 is the recent decision to make a grand transition from organic to regenerative farming, which has resulted thus far in 125 new regenerative beds spanning more than two kilometres. Another example is the almost complete removal of dairy products from the communal kitchen in EV2.

In addition to collective decisions that work to further sustainability, our empirical data also point to more subtle forms of agency with the potential to move practices in a more sustainable direction. While our data do not allow us to trace movement over time in people's behaviour, they do indicate that learning by example and participation can lead to altered behaviours like avoiding harmful materials, eating more sustainably and furthering biodiversity. An example of the latter is a project established by a small group in EV2 which photo-records all the insects spotted in the village and counts the number of species. Children particularly are engaged in this, and it seems likely that this and other biodiversity initiatives increase the awareness of and focus on biodiversity (see also Pisters et al., 2020, 2023).

Mechanisms, lifestyles and agency – some (meta)theoretical reflections

Following our analysis of the data and identification of the mechanisms above, we now turn to (meta)theoretical reflections. This involves a discussion of the work of Brand and Wissen (2021) in the light of critical realism and the research at hand. The reason for discussing this particular work is the close affinity between the mechanisms identified in this study and two main concepts in Brand and Wissen's analysis and the purpose of the comparison is to clarify the contribution of this article and the critical realist approach.

Brand and Wissen (2021) set out to understand normal everyday practices in the Global North and the power relations behind them. The most widespread mode of living they refer to as imperial. This mode of living revolves around high productive labour, high and growing consumption, cars and frequent flying (see also Schmelzer et al., 2022). Brand and Wissen uncover the close connection between the capitalist mode of production and ways of life, which marks capitalism today. Imperial life seems attractive to most people, or at least people find imperial life to be without alternatives. However, such a mode of living hides the destruction of humans and nature that it presupposes, thus being imperial. When the imperial mode of living expands by means of the global middle-class, crises and tensions worsen.

However, according to Brand and Wissen, the imperial mode of living opens the possibility of a transition towards a mode of living that is worth living for everyone without being destructive. Such a solidaric mode of living entails drastic reductions in resource use and CO₂ emissions and involves solidaric means of decision-making, co-existence and conflict resolution. They find such a mode of living to be exemplified by social movements, uprisings and alternative ways of living around the world. Still, for Brand and Wissen a clear path towards an eco-social transformation is yet to be seen.

The present study to some extent resembles Brand and Wissen's analysis, with middle-class ideals being like the imperial mode of living and the political economy of subsistence being similar to the solidaric mode of living. However, despite such similarities, the present study is in several ways distinct. Whereas Brand and Wissen embark on grand theory in the historical materialist tradition, making sweeping generalisations across global capitalism, the present paper studies sustainability and lifestyles in two Danish ecovillages by means of extensive empirical data. The present study aims to produce fallible and context-dependent knowledge. Beside this basic difference, other divergences can be identified. Whereas for Brand and Wissen the two modes of living seem to be mutually exclusive as absolute opposites, in the present study the two mechanisms exist in parallel. And whereas for Brand and Wissen modes of living are equal to actual ways of life, in the present study the mechanisms are ontologically distinct from the lifestyles they generate and sustain. For Brand and Wissen, it seems that prevailing ways of living can basically be deduced from the mode of production. In our approach, however, actual lifestyles involve *agency* and as such also decisions that cannot be predetermined or reduced to structures or modes (Buch-Hansen and Nielsen, 2024). Middle-class is not something *you are*, it is something *you do to some extent*.

By working deductively from the mode of production to the imperial mode of living and focusing almost exclusively on this mode, as well as emphasising that no overall alternative currently exists, Brand and Wissen can be said to contribute to naturalising capitalism and middle-class lifestyles in the Global North, as well as disregarding the actual diversity of political economies around the world. With Gibson-Graham (2006), we can say that by portraying capitalism and the corresponding mode of living as a global monolith, Brand and Wissen make the alternatives seem unrealistic and marginal. However, in our bottom-up analysis, the political economy of subsistence is by no means unreal or marginal in these villages.

Critical realism is of consequence here. Emphasising empirical data in conjunction with fallible in-depth abstraction and analyses and bringing in agency as ontologically non-reducible to structures and mechanisms is basic to critical realism (Buch-Hansen and Nielsen, 2020). Highlighting context and emphasising diversity and agency by having a bottom-up perspective and by using empirical data on social interactions and real-life experiences of people, are all together crucial in the present study. Critical realism provides precision when it comes to separating data, mechanisms and structures, and the methodology of abstraction can reveal specific deep factors shaping actual life while critical realists at the same time underline human agency (Buch-Hansen and Nielsen, 2024).

Concluding remarks

Degrowth scholarship on ecovillages and sustainability tends to be either theoretical or empirical. In either case, however, it most often highlights the exemplary nature of ecovillages when it comes to a degrowth transformation. Dissenting voices are few and marginal. In light of this, the present study contributes by reflecting critically and (meta)

theoretically on extensive empirical data in order to uncover and understand the nuances of the real-life experiences that unfold in two Danish ecovillages.

Applying critical realism, two crucial mechanisms regarding (un)sustainable lifestyles were identified in both villages: a subsistence political economy, and middle-class ideals. Both mechanisms are grounded in physical and social infrastructures in the villages. The political economy of subsistence generates and sustains more sustainable lifestyles and is in line with degrowth, whereas the opposite is the case with middle-class ideals.

Whereas subsistence political economies are particular to the villages, middle-class ideals are also much more widespread in Denmark, caused by grand structures of neo-liberal capitalism that work against degrowth. Importantly, lifestyles in the villages are very different and cannot be reduced to one mechanism or the other. Generally, and referring to the definition of sustainability and degrowth above, the villages are not sustainable, but some people in the villages have quite sustainable lifestyles that are compatible with a degrowth transformation. We have also found that over time, not only individual agency but also collective agency in the villages is paramount to sustainability, and that there are currently trends in the villages working both for and against a sustainable degrowth transformation.

Based on these findings, one might consider what can be done in the ecovillages to forward a degrowth transformation? Generally, it can be concluded that *more* subsistence political economy and *less* adherence to middle-class ideals would be steps in the right direction. Examples of such developments could play out on both the individual and the collective levels. When it comes to individual agency, examples include choosing not to fly, not owning a second home and consuming moderately. In terms of collective agency, examples include residents working together to increase regenerative farming on communal land and to promote biodiversity. Such actions are compatible with a sustainable degrowth transformation.

However, a deliberate transition towards more subsistence political economy and away from middle-class ideals through predefined village rules such as no flying, a low cap on consumption or a ten-hour work duty per week would in our analysis lead to less inclusivity in both villages. Given the direct democracy in the villages, it is difficult to see such rules pass through the decision-making process. There is in fact a tension between diversity and social cohesion, and the ability to attract and inspire many people on the one hand, and a more radical sustainability leading to deep conflicts and an exodus on the other hand. Thus, there is no straightforward route to sustainability and degrowth in these two ecovillages.

In addition, the analysis and findings presented in this paper also clearly illustrate that local solutions like ecovillages are not sufficient to curb the ongoing ecological catastrophe. Both middle-class lifestyles and ecovillages are embedded in larger unsustainable social and economic structures and in addition not everyone can live in ecovillages as this would give rise to unsustainable use of materials and land (Xue, 2014). Sustainable degrowth transitions therefore require changes at larger scales too, a suggestion that is abundant in the degrowth literature, as illustrated above. Examples of structural changes include global redistribution, a reduction of working hours and resource and emissions caps (Jackson, 2009; Kallis et al., 2020; Latouche, 2009). Importantly,

this is not a question of furthering solutions locally, nationally or internationally, but rather of pursuing multiscale solutions simultaneously. This is particularly the case, as there are interconnections between scales that can reinforce developments in both positive and negative directions when it comes to degrowth and sustainability. On a positive note, ecovillages can inspire sustainability transformations on other scales, while degrowth transformations on other scales can likewise increase sustainability in these two Danish ecovillages.

Acknowledgements

We would like to express our sincere gratitude to all the people in the two participating ecovillages who kindly took time to share their experiences and knowledge.

ORCID iDs

Peter Nielsen  <https://orcid.org/0009-0004-6566-1544>

Lone Riisgaard  <https://orcid.org/0000-0002-8827-3227>

Ethical approval

The project application which was approved by the Independent Research Fund Denmark contained detailed information about ethical considerations. Furthermore, the overall project has gone through ethical approval at Witwatersrand University in South Africa as they are a part of the project.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The data used in this paper were collected as part of the research project 'Producing alternative green futures: Exploring interconnections between green transitions and socioeconomic and political organisation' funded by the Independent Research Fund Denmark.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Informed consent

Informed consent was obtained verbally from all people interviewed.

References

- Bhaskar R (2008) *A Realist Theory of Science*. London: Routledge.
- Bhaskar R (2011) *Reclaiming Reality: A Critical Introduction to Contemporary Philosophy*. London: Routledge.
- Bhaskar R (2015) *The Possibility of Naturalism: A Philosophical Critique of the Contemporary Human Sciences*. London: Routledge.
- Brand U and Wissen M (2021) *The Imperial Mode of Living: Everyday Life and the Ecological Crisis of Capitalism*. London and New York: Verso.

- Buch-Hansen H (2019) Reorienting comparative political economy: From economic growth to sustainable alternatives. In: Chertkovskaya E, Paulsson A and Barca S (eds) *Towards a Political Economy of Degrowth*. London and New York: Rowman & Littlefield. pp. 39–53.
- Buch-Hansen H, Koch M and Nesterova I (2024) *Deep Transformations: A theory of Degrowth*. Manchester: Manchester University Press.
- Buch-Hansen H and Nesterova I (2021) Towards a science of deep transformations: Initiating a dialogue between degrowth and critical realism. *Ecological Economics* 190(December): 107188.
- Buch-Hansen H and Nesterova I (2023) Less and more: Conceptualising degrowth transformations. *Ecological Economics* 205(March): 107731.
- Buch-Hansen H and Nielsen P (2020) *Critical Realism: Basics and Beyond*. London: Bloomsbury.
- Buch-Hansen H and Nielsen P (2023) Critical realism, the climate crisis, and (de)growth. *Journal of Critical Realism* 22(3): 347–363.
- Buch-Hansen H and Nielsen P (2024) *Videnskabsteori i praksis*. Frederiksberg: Samfundslitteratur.
- Cattaneo C (2015) Eco-communities. In: D’Alisa G, Demaria F and Kallis G (eds) *Degrowth: A Vocabulary for a New Era*. Oxon and New York: Routledge, 165–168.
- Daly M (2017) Quantifying the environmental impact of ecovillages and cohousing communities: A systematic literature review. *Local Environment: The International Journal of Justice and Sustainability* 22(8): 1–20.
- Fletcher AJ (2017) Applying critical realism in qualitative research: Methodology meets method. *International Journal of Social Research Methodology* 20(2): 181–194.
- Fonseca RAA, Irving MDA and Nasri YXG, et al. (2022) Sustainability and social transformation: The role of ecovillages in confluence with the *pluriverse* of community-led alternatives. *Climate Action* 1(1): 23.
- Gausset Q (2020) Stronger together: How Danish environmental communities influence behavioural and societal changes. In: Hoff J, Gausset Q and Lex S (eds) *The Role of Non-State Actors in the Green Transition: Building a Sustainable Future*. Oxon and New York: Routledge, 52–70.
- Gausset Q and Jensen PD (2024) Living sustainably in a Danish eco-community: How social and physical infrastructure affect carbon footprints. *NPJ Climate Action* 3(1): 33.
- Gibson-Graham JK (2006) *The End of Capitalism (As We Knew It): A Feminist Critique of Political Economy*. Minneapolis: University of Minnesota Press.
- Hoff J, Gausset Q and Lex S (2020) Introduction. In: Hoff J, Gausset Q and Lex S (eds) *The Role of Non-State Actors in the Green Transition: Building a Sustainable Future*. Oxon and New York: Routledge, pp.1–13.
- Jackson T (2009) *Prosperity Without Growth: Economics for a Finite Planet*. London and New York: Earthscan.
- Kallis G, Paulson S and D’Alisa G, et al. (2020) *The Case for Degrowth*. Cambridge and Medford: Polity Press.
- Latouche S (2009) *Farewell to Growth*. Cambridge: Polity Press.
- Litfin KT (2014) *Ecovillages: Lessons for Sustainable Community*. Cambridge and Malden: Polity Press.
- Lockyer J (2017) Community, commons, and degrowth at Dancing Rabbit Ecovillage. *Journal of Political Ecology* 24(1): 519–542.

- Marckmann B, Gram-Hansen K and Christensen TH (2012) Sustainable living and co-housing: Evidence from a case study of ecovillages. *Built Environment* 38(3): 413–429.
- Mies M and Shiva V (2013) *Ecofeminism*. London and New York: Zed Books.
- Mocca E (2020) The local dimension in the degrowth literature: A critical discussion. *Journal of Political Ideologies* 25(1): 78–93.
- Nguyen T (2019) *The production of community in Munksøgård: Community, values and infrastructures*. PhD Thesis, University College London, UK.
- Nielsen P (2011) *Velstandssamfundet*. Frederiksberg: Frydenlund.
- Næss P (2006) Unsustainable growth, unsustainable capitalism. *Journal of Critical Realism* 5(2): 197–227.
- Parker J (2007) Sustainability. In: Hartwig M (ed) *A Dictionary of Critical Realism*. London and New York: Routledge, pp.449–451.
- Piketty T (2020) *Capital and Ideology*. Cambridge, MA and London: The Belknap Press of Harvard University Press.
- Pisters SR, Vihinen H and Figueiredo E (2020) Inner change and sustainability initiatives: Exploring the narratives from eco-villagers through a place-based transformative learning approach. *Sustainability Science* 15(2): 395–409.
- Pisters SR, Vihinen H and Figueiredo E, et al. (2023) We learned the language of the tree: Ecovillages as spaces of place-based transformative learning. *Journal of Transformative Education* 21(1): 59–83.
- Polewsky M, Hankammer S and Kleer R, et al. (2024) Degrowth vs. green growth: A computational review and interdisciplinary research agenda. *Ecological Economics* 217(March): 108067.
- Schmelzer M, Vetter A and Vansintjan A (2022) *The Future Is Degrowth: A Guide to a World beyond Capitalism*. London and New York: Verso.
- Temesgen AK (2020) Building an island of sustainability in a sea of unsustainability? A study of two ecovillages. *Sustainability* 12(24): 10585.
- Tilsted JP and Bjørn A (2023) Green frontrunner or indebted culprit? Assessing Denmark's climate targets in light of fair contributions under the Paris Agreement. *Climatic Change* 176(8): 03.
- Trainer T (2006) On eco-villages and the transition. *The International Journal of Inclusive Democracy* 2(3).
- Trainer T (2012) De-growth: Do you realise what it means? *Futures* 44(6): 590–599.
- Vandeventer JS and Schmid B (2024) What does degrowth do in/to empirical research? Methodological deliberations on placing degrowth 'in the world'. *Ecological Economics* 221(July): 108206.
- Verco N (2019) Christiania: A poster child for degrowth? In: Nelson A and Schneider F (eds) *Housing for Degrowth: Principles, Models, Challenges and Opportunities*. Oxfordshire and New York: Routledge, 99–108.
- Vogel J and Hickel J (2023) Is green growth happening? An empirical analysis of achieved versus Paris-compliant CO₂-GDP decoupling in high-income countries. *The Lancet Planetary Health* 7(9): e759–e769.
- Windegger F and Spash CL (2023) Reconceptualising freedom in the 21st century: Neoliberalism vs. degrowth. *New Political Economy* 28(4): 554–573.
- Xue J (2014) Is eco-village/urban village the future of a degrowth society? An urban planner's perspective. *Ecological Economics* 105(September): 130–138.